

Mastermind behind the watchful eye!

It took two-three years of extensive research and development by a team of five experts at ideaForge to develop Netra. "We began with the development of autopilot, which ended up being the world's smallest and lightest autopilot. There have been challenges at each and every stage throughout the development, right from its first flight till date. And every failure has been an essential part of strengthening our base," recalls Mehta.

stand harsh conditions like wind and poor visibility. "We have studied wind effects on the vehicle and done a lot of aerodynamic design changes for improved wind resistance," says Mehta. Further, it can be operated at night and soon even indoors.

"The thermal camera option allows Netra to operate at night, in fog or any other kind of poor visibility condition. Since the vehicle is completely autonomous, the user need not worry about the navigation issues in such conditions," says Mehta.

Smart auto pilot. The level of autonomy in Netra is higher than in any other UAV as it allows hands-free operation after the operator commands the vehicle to go to a specified destination. "The software displays maps (satellite imagery from Google maps) around the current location of the vehicle along with trajectory of the moving vehicle. For navigation, the user just needs to mark a way-point on the map. Double-clicking the point will command the vehicle to start moving towards that point," says Mehta.

The BlueFire Autopilot, which is the brain of Netra, takes on thereafter. The autopilot controller takes over the tasks of take-off and navigation as

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per the flight plan, at every instance providing the corrections and auto stabilising the platform due to inherent motion and wind gusts. This flight plan can be dynamically changed during flight, providing high levels of autonomy to the operator.

"The intelligent autopilot communicates with the ground station, gathers data from all the sensors, stabilises the vehicle and navigates it. It has been developed in-house by ideaForge and is world's smallest and lightest autopilot available promising to relieve the operator from skilful manoeuvring," says Mehta. Further, built-in intelligence of the controller allows failsafe operation in case of communication failure with base station or low battery.

Foolproof safety plan. Netra can fly for half an hour on a single battery charge, thanks to use of high-quality Li-Po batteries having the highest energy density (energy/weight ratio). Even if the battery gives way, Netra has several levels of fail-safe modes like low-battery fail-safe and communication-loss fail-safe. In the case of low battery, the UAV automatically comes back and lands by itself. During a communication failure between the vehicle and ground control station, Netra starts coming back and when it regains communication, it hovers and waits for the next command.

"To ensure safe operation, the ve-

hicle keeps track of the battery voltage and signal strength. Whenever these two reach a critical value, it triggers the fail-safe mode and the vehicle starts coming back," says Mehta.

Every element of data security has been taken care of. "Users have the option to switch over to a licenced frequency band which is not easily accessible by a third person. This will be a hindrance against any meddling. Soon we will give the option of digital transmission that would be encrypted, making it impossible to tap," says Mehta.

Very portable. Netra is extremely small, lightweight and portable, requiring only two men to carry the system along with the control station to field locations. It weighs only 1.5 kg and is 90x90 cm² in size.

"Weight is of prime importance in case of flying vehicles. We did many optimisations in every component we used. Further, the body is made of carbon fibre, which is a very light and strong material," says Mehta.

Peek into the future

Looking ahead, ideaForge is determined to make continuous improvements in Netra. "In the immediate future, better communication and improved image intelligence are the areas that we wish to focus on. We are also working on reaching out to potential customers. We are targeting armed forces, paramilitary forces, special task forces, ATS, security agencies, fire fighters, disaster management groups, etc and have already started holding demonstrations for them," says Mehta, on an optimistic note. ●

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